

Tung Yan Zuo, Darren

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Personal Website: darren-tung.github.io/Profile/

PROFESSIONAL SUMMARY

Software engineer with two years of internship experience developing full-stack, mobile, cloud, and automation solutions. Proficient in TypeScript, Python, React, React Native, AWS, Azure, and Docker, with a proven track record of improving software scalability and operational efficiency. Collaborative and adaptable developer experienced in Agile environments, team leadership, and building reliable solutions for diverse applications.

EDUCATION

National University of Singapore

Bachelor of Computing in Computer Science (Hons)

- **Specialisation:** Software Engineering & Networking and Distributed Systems

Aug 2023 - Jan 2027

Singapore, SG

University of Illinois Chicago

Entrepreneurship curriculum under NUS Overseas College (NOC)

Aug 2025 - Jul 2026

United States, Illinois, Chicago

Ngee Ann Polytechnic

Diploma in Engineering Science

- **GPA:** 3.9266 / 4.0
- **Awards:** DSTA Polytechnic Engineering Scholarship, Director's List (Year 2.1), Director's List (Year 1.2)

Apr 2018 - May 2021

Singapore, SG

PROFESSIONAL EXPERIENCE

Rhaeos

Software Engineering Intern

- Re-architected entire **Bluetooth Low Energy (BLE)** Manager Submodule infrastructure using **TypeScript, AWS & React Native** to facilitate seamless integration, scalability, and usage within all Rhaeos mobile applications
- Transitioned core product from its predecessor to a clinically ready state utilising **TypeScript, AWS, NodeJS, React Native, XCode & Python** directly supporting 2+ clinical studies & its path to FDA 510k approval
- Created the Validator App using **TypeScript, React Native, & BLE** to verify hardware functionality which established a robust **QA** protocol projected to save hundreds of hours of manual testing
- Architected and built foundational framework for the Rhaeos OTP forgot-password system utilising **AWS (Lambda, Cognito, Simple Email System, EventBridge, CloudTrail)**

Aug 2025 - Jul 2026

United States, Illinois, Chicago

Y3 Technologies

Software Engineering Intern

- **Led automation** project to convert **Jira Tickets** to carefully curated test cases, which will be converted to Excel files to be sent to testing team, using **Python, Anthropic AI & Prompt Engineering**
- Set up & facilitated **CI/CD deployment pipeline** to company's server utilising **GitHub Actions, Bash, Git & Docker**
- Created a dynamic **UI** for non-developers to interact with core functionality, minimising interaction with **CLI** using **Vite, Tailwindcss, ReactJS, Typescript & Shadcn**, while having a separate administrator window to view logs
- Projected to impact **over 100** staff members & **save Y3 Technologies over \$10000** annually

May 2025 - Aug 2025

Singapore, SG

Backbone

Full Stack Software Engineering Intern

- Led full-stack redevelopment of Backbone's corporate website utilising **Next.js, Strap, Shadcn, TypeScript, Docker & Tailwind CSS**, delivering enhanced performance, maintainability, & consistent design across all pages with **CI/CD pipeline** via **GitHub Actions** for automated deployment to **Microsoft Azure**
- Engineered **Python** automation solution that converts client financial statements into dynamic business rules stored in **MySQL** database, reducing manual processing time by **80%** & **eliminating data entry errors**
- Architected seamless data export pipeline to **Microsoft Azure**, enabling automatic integration of custom rule sets with Backbone's core platform & delivering personalised client solutions with zero manual intervention

Sep 2024 - May 2025

Singapore, SG

TECHNICAL SKILLS

- **Programming Languages:** C, C++, C#, Python, JavaScript, HTML, Java, R, TypeScript
- **Frameworks & Technologies:** ReactJS, NodeJS, WebRTC, React Native, OpenCV, Shadcn, Vite, NextJS
- **Databases & Cloud Services:** MySQL, MongoDB, Firebase, Microsoft Azure, Amazon Web Services, SQLancer
- **Tools & DevOps:** Git, GitHub Actions, Docker, Google App Scripts, Microsoft VBA, R Studios, XCode

CERTIFICATION

Amazon Web Services (AWS) Certified AI Practitioner Certificate	Mar 2026
Amazon Web Services (AWS) Certified Solution Architect - Associate	Aug 2025
Scrum Alliance Certified ScrumMaster (CSM)	Jun 2025
Amazon Web Services (AWS) Certified Cloud Practitioner Certificate	Jun 2025
WILEY Certified Data Analyst	Apr 2022

PROJECTS

CartOpix	Aug 2026 - Present
NUS Software Engineering Project	
• Collaborate with 6 developers to develop a Cloud-Based Grocery Basket Optimisation SaaS project using Next.js, Shadcn, TypeScript, ReactJS, NodeJS & AWS (S3, DynamoDB, Cognito, Lambda, EventBridge, CloudTrail)	
SQLancer Benchmark	Jan 2025 - May 2025
NUS Software Engineering Project	
• Led a team of 3 developers to design & implement a web app using Next.js, Shadcn, TypeScript, ReactJS & NodeJS, which will allow non-developers to use our product with no issues • Collaborated with Metrics team & facilitated integration of frontend & backend processes with unit & integration testing, Github Actions, CI-CD Pipelining, SQLancer & bash scripts	
TalentSG Web App	Oct 2024 - Dec 2024
NUS Software Engineering Project	
• Designed web app on Canva & integrated it utilising JavaX, CSS, CI-CD Pipelining & Java • Led a team of 5 developers & conducted weekly sprints & scrum meetings while leveraging Agile & Git	
Health Screening 2024	May 2024 - Aug 2024
Public Health Service Project (Supported by NUS Medicine & NUS Medical Society)	
• Developed & updated web-based queue system & data storage with JavaScript, ReactJS & MongoDB • Led a team of 4 developers & streamlined its development leveraging Agile & Git, as well as, communicating with the medical PHS committee & delegating objectives for team members • Impacted over 1000 volunteers & participants who underwent a health screening by PHS	
UniFyTime	May 2024 - Aug 2024
Independent Software Development Project (Orbital - Advance)	
• Worked in pairs to propose & develop a web-based calendar system using JavaScript, ReactJS & Firebase API • Designed user interface (UI) for optimum user experience (UX), structuring information architecture for intuitive navigation aligning with modern designs & best practices, real-time synchronisation with Firebase & Firestore • Coordinated effective communication within team through daily developmental sprints utilising Agile & Git • Encompassed Telegram Bot Integration using JavaScript & Telegraf.js	

LEADERSHIP EXPERIENCE

Treasurer (Grassroots Leader)	Oct 2016 - Present
Zhenghua Community Club Youth Network (Community Service)	
• Oversee claims & expenditures, approve reimbursements, & ensure accurate financial documentation • Review SOA & transactions before meetings, maintaining transparency & accountability in committee funds • Organised numerous community events that reached out to over 300 residents in the Zhenghua community • Volunteered in various community events, such as a weekly food rescue event • Awards: People's Association Certificate of Appreciation (5 Years Long Service Award)	